

LESSON STRUCTURE

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2.0 Objective

In this unit in addition to Bilinear Transformations, Cross ratio, normal form of Bilinear Transformations will be discussed. Special emphasis will be given on conformal mapping, special bilinear transformation, the mapping, $W = z^n, W = z^2$ and the inverse mapping $W = z^{1/2}$. At the end extended complex plane will be discussed.

2.1 Introduction

Before going to discuss Bilinear Transformation, we shall define

Bilinear Transformation

Transformation or mapping, conformal transformation and linear transformation.

The equation $\left. \begin{matrix} u = u(x, y) \\ v = v(x, y) \end{matrix} \right\} \dots\dots (1)$ define a transformation or mapping by means

of which a correspondence between uv plane and xy -plane can be established. If corresponding to each point of xy -plane, there is a unique point of uv -plane and conversely, then such a transformation is called **one-one transformation**. The corresponding points in the two planes are called images of each other. Hence, by means of equations (1), a region or curve of xy - plane is said to transform or mapped on or represented by the corresponding curve of the uv -plane.

The linear transformation : A transformation of the form $w = az + b$, is called a linear transformation, where a and b are complex constants.

2.2 Bilinear Transformation or Mobius transformation :

A transformation of the form $w = \frac{az + b}{cz + d} \dots\dots (1)$ is called a Bilinear transformation of linear fractional transformation, where a, b, c, d are complex constants.

Such type of transformation was first studied by Mobius and hence it is sometimes called mobius transformation.

Equation (1) may be derived as $CW_z + Wd - az - b = 0$ which is clearly a linear both in W and Z . That is why, it is called Bilinear. Here we assume that $ad - bc \neq 0$ which is called the determinant of the transformation. The transformation (1) is called **normalized** if $ad - bc = 1$ / if the determinant vanishes, then w is merely a constant as shown below.

Let W_1 and W_2 be two values of W corresponding to Z_1 and Z_2 in (1), then

$$\begin{aligned} W_2 - W_1 &= \frac{az_2 + b}{cz_2 + d} - \frac{az_1 + b}{Cz_1 + d} \\ &= \frac{acz_1z_2 + bc z_1 + az_2d + bd - acz_1z_2 - bc z_2 - az_1d - bd}{(Cz_1 + d)(Cz_2 + d)} \\ &= \frac{z_2(ad - bc) - z_1(ad - bc)}{(Cz_1 + d)(Cz_2 + d)} \end{aligned}$$

$$= \frac{(z_2 - z_1)(ad - bc)}{(Cz_1 + d)(Cz_2 + d)} = 0 \text{ if } ad - bc = 0$$

$\therefore W_2 - W_1 = 0$ if $ad - bc = 0$ This shows that W is a constant

2.2.1 Critical Points : Let

$$w = \frac{az + b}{Cz + d} \quad \dots\dots\dots (1)$$

$$\therefore z = \frac{b - wd}{wc - a} \quad \dots\dots\dots (2)$$

From (1) $\frac{dw}{dz} = \frac{ad - bc}{(cz + d)^2}$

This means $\frac{dw}{dz} = \infty$ if $z = -d/c$

and $\frac{dw}{dz} = 0$ if $z = \infty$

The points $z = -d/c, z = \infty$ are called critical points where the conformal property does not hold good.

Also from (2) it is clear that for each $z \neq -d/c$ we have a value of W and for each $W \neq a/c$ there corresponds a value of Z and the correspondence between W and Z is one-one. The exceptional points $Z = -d/c$ and $W = a/c$ are mapped into the points $W = \infty$ and $Z = \infty$ respectively.

2.2.2 Extended complex plane : These points will not remain exceptional if we adjoin a new point called point at infinity denoted by ∞ to the complex plane and the complex plane in this case is called extended complex plane. Thus the critical point $Z = \infty$ corresponds to the point $W = a/c$.

Hence we can now say that the bilinear sets up a one-one correspondence between the points of entire extended complex z -plane upon the entire extended w -plane.

2.3 Resultants or Product of two bilinear transformations

Consider two bilinear transformation

$$W = \frac{az + b}{cz + d} \text{ such that } ad - bc \neq 0 \quad \dots\dots\dots (1)$$

$$\text{and } \xi = \frac{a_1 w + d_1}{c_1 w + d_1} \text{ such that } a_1 d_1 - b_1 c_1 \neq 0 \quad \dots\dots\dots (2)$$

Putting the value of W from (1) in equation (2), we get

$$\xi = \frac{a_1 \left(\frac{az + b}{cz + d} \right) + d_1}{c_1 \left(\frac{az + b}{cz + d} \right) + d_1} = \frac{(aa_1 + b_1 c)z + (b_1 d + a_1 b)}{(c_1 a + d_1 c)z + (d_1 d + c_1 b)}$$

$$\text{Writing } A = aa_1 + b_1 c, \quad B = b_1 d + a_1 b, \quad C = c_1 a + d_1 c, \quad D = d_1 d + c_1 b$$

$$\text{We get } \xi = \frac{AZ + B}{CZ + D}$$

$$\begin{aligned} \text{Here } AD - BC &= (aa_1 + b_1 c)(d_1 d + c_1 b) - (b_1 d + a_1 b)(c_1 a + d_1 c) \\ &= aa_1 d_1 d + bb_1 c c_1 - b_1 c_1 a d - a_1 d_1 b c \\ &= (a_1 d_1 - b_1 c_1)(ad - bc) \neq 0 \text{ by (1) \& (2)} \end{aligned}$$

$$\text{Thus } \xi = \frac{AZ + B}{CZ + D} \text{ such that } AD - BC \neq 0$$

This is a bilinear transformation. This bilinear transformation is called the resultant or product of the transformation resultant (1) and (2). Thus we may say bilinear transformation forms a group.

2.3.1 Geometrical Inversion : Every bilinear transformation is the resultant of bilinear transformation with simple geometrical imports.

Consider the bilinear transformation

$$w = \frac{az + b}{cz + d} \quad \dots\dots\dots (1)$$

Where $ad - bc \neq 0, c \neq 0$

$$\text{from (1) } w = \frac{a}{c} \cdot \frac{z + \left(\frac{b}{a}\right)}{z + \left(\frac{d}{c}\right)} = \frac{a}{c} \left[1 + \frac{\frac{b}{a} - \frac{d}{c}}{z + \frac{d}{c}} \right]$$

$$w = \frac{a}{c} + \frac{bc - ad}{c^2} \cdot \frac{1}{z + \frac{d}{c}}$$

$$\text{Putting } Z_1 = z + \frac{d}{c}, Z_2 = \frac{1}{Z_1}, Z_3 = \frac{bc - ad}{c^2} \cdot Z_2$$

We obtain $w = \frac{a}{c} + Z_3$ which is similar to $Z_1 = z + \frac{d}{c}$.

The above three auxiliary transformations namely Z_1, Z_2, Z_3 are of the form

$$W = Z + \alpha, W = \frac{1}{Z}, W = \beta Z$$

This proves that every general bilinear transformation is the resultant of the bilinear transformations

$$W = Z + \alpha, W = \frac{1}{Z}, W = \beta Z$$

Where (i) $W = Z + \alpha$ represents translation

(ii) $W = \frac{1}{Z}$ represents inversion

and (iii) $W = \beta Z$ represents dilation.

2.4 Cross Ratio

If Z_1, Z_2, Z_3, Z_4 are distinct points, then the ratio $\frac{(Z_4 - Z_1)(Z_2 - Z_3)}{(Z_2 - Z_1)(Z_4 - Z_3)}$ is called

the cross ratio of Z_1, Z_2, Z_3, Z_4 . This ratio is invariant under the bilinear transformation.

Number of four distinct cross-ratio :

From four points Z_1, Z_2, Z_3 and Z_4 lying in the Z -plane, we can obtain different cross ratios according to the order in which the points are taken. Since the four points can permute themselves in only six-distinct cross ratios. This is so

Bilinear Transformation

because if we inter change any points and then interchange remaining two, the cross ratios of the points in this new order will be the same. In fact, we have

$$(Z_1, Z_2, Z_3, Z_4) = (Z_2, Z_1, Z_4, Z_3) = (Z_3, Z_4, Z_1, Z_2) = (Z_4, Z_3, Z_2, Z_1) \dots\dots\dots (1)$$

$$(Z_1, Z_2, Z_4, Z_3) = (Z_2, Z_1, Z_3, Z_4) = (Z_3, Z_4, Z_2, Z_1) = (Z_4, Z_3, Z_1, Z_2) \dots\dots\dots (2)$$

$$(Z_1, Z_3, Z_2, Z_4) = (Z_2, Z_4, Z_1, Z_3) = (Z_3, Z_1, Z_4, Z_2) = (Z_4, Z_2, Z_3, Z_1) \dots\dots\dots (3)$$

$$(Z_1, Z_3, Z_4, Z_2) = (Z_2, Z_4, Z_3, Z_1) = (Z_3, Z_1, Z_2, Z_4) = (Z_4, Z_2, Z_1, Z_3) \dots\dots\dots (4)$$

$$(Z_1, Z_4, Z_2, Z_3) = (Z_2, Z_3, Z_1, Z_4) = (Z_3, Z_2, Z_4, Z_1) = (Z_4, Z_1, Z_3, Z_2) \dots\dots\dots (5)$$

$$(Z_1, Z_4, Z_3, Z_2) = (Z_2, Z_3, Z_4, Z_1) = (Z_3, Z_2, Z_1, Z_4) = (Z_4, Z_1, Z_2, Z_3) \dots\dots\dots (6)$$

2.4.1 The cross ratio (Z_1, Z_2, Z_3, Z_4) is real iff the four points Z_1, Z_2, Z_3, Z_4 lie on a circle or on a straight line.

Proof : By definition of cross ratio

$$(Z_1, Z_2, Z_3, Z_4) = \frac{(Z_4 - Z_1)(Z_2 - Z_3)}{(Z_4 - Z_3)(Z_2 - Z_1)} = \frac{(Z_4 - Z_1)/(Z_4 - Z_3)}{(Z_2 - Z_1)/(Z_2 - Z_3)}$$

$$\Rightarrow \arg(z_1, z_2, z_3, z_4) = \arg\left(\frac{z_4 - z_1}{z_1 - z_3}\right) - \arg\left(\frac{z_2 - z_1}{z_2 - z_3}\right)$$

$= 0$ or $\pm \pi$ if points lie on a circle or straight line

2.5 Preservance of Cross Ratio :

Prove that cross ratio remains invariant under a bilinear transformation.

Proof : Let w_1, w_2, w_3, w_4 be the images of four distinct points z_1, z_2, z_3, z_4 respectively under a bilinear transformation.

$$w = \frac{az + b}{cz + d} \text{ where } ad - bc \neq 0 \dots\dots\dots (1)$$

If we show that

$$\frac{(w_4 - w_1)(w_2 - w_3)}{(w_2 - w_1)(w_4 - w_3)} = \frac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)} \dots\dots\dots (2)$$

The result will follow from (1)

$$\begin{aligned} W_r - W_s &= \frac{az_r + b}{cz_r + d} - \frac{az_s + b}{cz_s + d} \\ \Rightarrow W_r - W_s &= \frac{(ad - bc)(z_r - z_s)}{(cz_r + d)(cz_s + d)} \dots\dots\dots (3) \end{aligned}$$

$$\begin{aligned} \text{This } \Rightarrow (w_4 - w_1)(w_2 - w_3) &= \frac{(ad - bc)(z_4 - z_1)}{(cz_4 + d)(cz_1 + d)} \cdot \frac{(ad - bc)(z_2 - z_3)}{(cz_2 + d)(cz_3 + d)} \\ \Rightarrow (w_4 - w_1)(w_2 - w_3) &= \frac{(ad - bc)^2 (z_4 - z_1)(z_2 - z_3)}{(cz_1 + d)(cz_2 + d)(cz_3 + d)(cz_4 + d)} \end{aligned}$$

Similarly again from (3) prove that

$$(w_2 - w_1)(w_4 - w_3) = \frac{(ad - bc)^2 (z_2 - z_1)(z_4 - z_3)}{(cz_1 + d)(cz_2 + d)(cz_3 + d)(cz_4 + d)}$$

Dividing the last two results we get the required equation (2).

2.6 Fixed Point of a Bilinear Transformation :

To prove that in general there are two values of Z (invariant points) for which $w=z$, but there is only one if

$$(a - d)^2 + 4bc = 0$$

Show that if there are distinct invariant points p and q , the transformation may be put in the form (normal form)

$$\frac{w - p}{w - q} = \frac{k(z - p)}{(z - q)} \text{ and that if, there is only one invariant points } p, \text{ the}$$

transformation may be put in the form (normal form) $\frac{1}{w - p} = \frac{1}{z - p} + k$.

Proof : Let us consider the bilinear transformation $w = \frac{az + b}{(cz + d)}$ (1)

(i) The invariant points are given by

$$w = z$$

$$\text{i.e. } z = \frac{az + b}{cz + d}$$

this gives $cz^2 - (a - d)z - b = 0$ (2)

Solving
$$Z = \frac{(a - d) \pm \sqrt{(a - d)^2 + 4bc}}{2c}$$

$$\left. \begin{array}{l} \text{Taking } p = \frac{(a - d) + \sqrt{(a - d)^2 + 4bc}}{2c} \\ \text{and } q = \frac{(a - d) - \sqrt{(a - d)^2 + 4bc}}{2c} \end{array} \right\} \text{ (3)}$$

This shows that p, q are distinct points and are invariant points for the transformation (1).

If $(a - d)^2 + 4bc = 0$, then (3) deduce that

$$p = q = \frac{a - d}{2c} \text{ (4)}$$

Hence there is only one invariant point namely $p = \frac{a - d}{2c}$ if $(a - d)^2 + 4bc = 0$

(ii) Suppose p and q are distinct invariant points the transformation (1), so that they satisfy (2). As a result of which

$$Cp^2 - (a - d)p - b = 0$$

$$Cq^2 - (a - d)q - b = 0$$

for which
$$\left[\begin{array}{l} Cp^2 - ap = b - pd \\ Cq^2 - aq = b - qd \end{array} \right] \text{ (5)}$$

$$\begin{aligned}
 \frac{w-p}{w-q} &= \frac{\left(\frac{az+b}{cz+d}\right)-p}{\left(\frac{az+b}{cz+d}\right)-q} = \frac{(az+b)-pcz-pd}{(az+b)-cqz-qd} \\
 &= \frac{(a-pc)z+b-pd}{(a-cq)z+b-qd} = \frac{(a-pc)z+cp^2-ap}{(a-cq)z+cq^2-aq} \text{ using (5)} \\
 &= \frac{(a-cp)(z-p)}{(a-cq)(z-q)}
 \end{aligned}$$

Taking $k = \frac{a-cp}{a-cq}$, we get $\frac{w-p}{w-q} = k \left(\frac{z-p}{z-q} \right)$ (6) which is required.

(iii) Let there be only one invariant point p , so that

$$p = \frac{a-d}{2c}, \quad cp^2 - ap = b - pd \text{ by (4) and (5)}$$

$$\begin{aligned}
 \text{Now } \frac{1}{w-p} &= \frac{1}{\frac{az+b}{cz+d}-p} = \frac{cz+d}{(a-cp)z+b-pd} \\
 &= \frac{cz-(2cp-a)}{(a-cp)z+cp^2-ap} = \frac{(cz-cp)+a-cp}{(a-cp)(z-p)} \\
 &= \frac{c(z-p)}{(a-cp)(z-p)} + \frac{a-cp}{(a-cp)(z-p)} \\
 &= \frac{c}{a-cp} + \frac{1}{z-p}
 \end{aligned}$$

Taking $k = \frac{c}{a-cp}$, we get $\frac{1}{w-p} = k + \frac{1}{z-p}$ which is required

This completes the proof.

Bilinear Transformation

(iv) Every bilinear transformation maps circles or straight lines into circles or straight lines.

or

In a bilinear transformation, a circle transforms into a circle and inverse points transform into inverse points. In the particular case in which the circle becomes a straight line, inverse points become symmetrical about the line.

or

Find the condition that the transformation $w = \frac{az + b}{cz + d}$ transforms the unit circle in w-plane into straight line of z-plane.

Proof : The equation of a circle with p and q as inverse points in z-plane is given by

$$\left| \frac{z-p}{z-q} \right| = k, \quad k \neq 1 \quad \dots\dots\dots (1)$$

(In case of a straight line $k=1$)

The bilinear transformation $w = \frac{az + b}{cz + d}$

So that $z = \frac{dw - b}{-cw + a} \dots\dots\dots (2)$

transforms the equation (1) into the form

$$\left| \frac{\frac{dw-b}{-cw+a} - p}{\frac{dw-b}{-cw+a} - q} \right| = k$$

or
$$\left| \frac{w(d+cp) - (ap+b)}{w(d+cq) - (aq+b)} \right| = k$$

or
$$\left| \frac{w-p'}{w-q'} \right| = \left| \frac{d+cq}{d+cp} \right| \cdot k$$

Where $p' = \frac{ap+b}{cp+d} = \left[\frac{az+b}{cz+d} \right]_{z=p}$

$$q' = \frac{aq+b}{cq+d} = \left[\frac{az+b}{cz+d} \right]_{z=q}$$

Thus $\left| \frac{w-p'}{w-q'} \right| = k'$

evidently $k' \neq 1$ even if $k = 1$

Clearly the transformed equation (3) represents a circle in w-plane with p' and q' as inverse points. In particular case when $k=1$, (1) represents a straight line which bisects perpendicularly the line joining p to q . The bilinear transformation transforms the line (1) of z-plane into a circle in w-plane and the points p and q symmetrical about the line becomes inverse point in w-plane.

Note : The above theorem may be read as *Obtain an expression which can be used to transform one given circle into another given circle or straight line.*

2.7 Some special Cases of Bilinear Transformations.

2.7.1 Find all the Mobius (linear) transformation which transforms the half plane $I(z) > 0$ into circles $|w| \leq 1$.

Proof : Suppose that the bilinear transformation $w = \frac{az+b}{cz+d}$ (1)

transforms the upper half plane $I(z) = y \geq 0$, into the unit circle $|w| \leq 1$.

(1) is expressible as $w = \frac{a}{c} \cdot \frac{z + (b/a)}{z(d/c)}$ (2)

This $\Rightarrow c \neq 0$, otherwise the point at infinity in the two planes would correspond.

If $w = 0$, then from (2) $\Rightarrow z = -b/a$

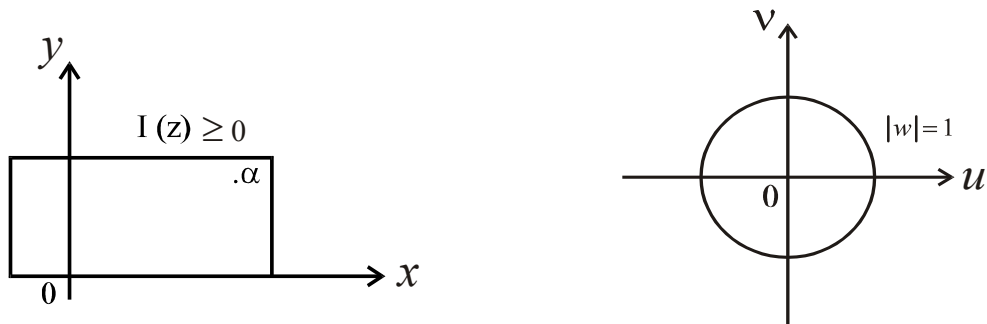


Figure - I

Bilinear Transformation

If $w = \infty$ then (2) $\Rightarrow z = -d/c$

Also $w = 0$, $w = \infty$ are the inverse points of $|w| = 1$ we have seen that the transformation (2) transforms a straight line of z -plane into a circle and points symmetrical about the line transform into inverse points of the circle w -plane. Hence the points $z, \bar{z}$ symmetrical about the real axis i.e. $I(z) = 0$ will correspond to $w = 0$, $w = \infty$ respectively. Hence we may take

$$-b/a = \alpha, -d/c = \bar{\alpha}$$

Then (2) becomes $w = \frac{a}{c} \left(\frac{z - \alpha}{z - \bar{\alpha}} \right)$ (3)

The point $z = 0$ on the boundary of the half plane $I(z) \geq 0$ must correspond to a point on the boundary of the circle $|w| = 1$ so that

$$1 = |w| = \left| \frac{a}{c} \right| \left| \frac{0 - \alpha}{0 - \bar{\alpha}} \right| = \left| \frac{a}{c} \right| \text{ as } |\alpha| = |\bar{\alpha}|$$

This $\Rightarrow \frac{a}{c} = e^{i\lambda}$ where λ is real

$$w = e^{i\lambda} \left(\frac{z - \alpha}{z - \bar{\alpha}} \right)$$
 (4)

evidently $z = \alpha$ gives $w = 0$. But $w = 0$ is an interior point of the circle $|w| = 1$.

Hence $z = \alpha$ must be a point of the upper half plane, i.e. $I(\alpha) > 0$. With this condition (4) is the required transformation.

Verification $w \bar{w} = 1 \frac{z - \alpha}{z - \alpha} \cdot e^{i\lambda} \frac{\bar{z} - \bar{\alpha}}{\bar{z} - \bar{\alpha}} e^{-i\lambda} = 1$

$$= \frac{(z - \alpha)(\bar{z} - \bar{\alpha})}{(z - \bar{\alpha})(\bar{z} - \alpha)} = 1$$

or
$$w \bar{w} - 1 = \frac{z\bar{z} - z\bar{\alpha} - \alpha\bar{z} + \alpha\bar{\alpha} - (z\bar{z} - z\alpha - \bar{\alpha}\bar{z} + \alpha\bar{\alpha})}{(z - \bar{\alpha})(\bar{z} - \alpha)}$$

$$= \frac{z(\alpha - \bar{\alpha}) - \bar{z}(\alpha - \bar{\alpha})}{|z - \bar{\alpha}|^2} = \frac{(\alpha - \bar{\alpha})(z - \bar{z})}{|z - \bar{\alpha}|^2}$$

$$= \frac{2i I(\alpha).2i I(z)}{|z - \bar{\alpha}|^2} = \frac{4 I(\alpha).I(z)}{|z - \bar{\alpha}|^2}$$

finally,
$$\frac{4 I(\alpha).I(z)}{|z - \bar{\alpha}|^2} = 1 - |w|^2 \dots\dots\dots (5)$$

Also we have proved that $I(\alpha) > 0$

Now from (5), it is clear that

(i) $I(z) = 0$ corresponds to $1 - |w|^2 = 0$ i.e. $|w| = 1$

(ii) $I(z) > 0$ corresponds to $1 - |w|^2 > 0$ i.e. $|w| < 1$

finally, $I(z) > 0$ corresponds to $|w| \leq 1$ by virtue of transformation (4)

2.7.2. Find the Bilinear Transformation which transforms $R(z) \geq 0$ into the unit circle $|w| \leq 1$.

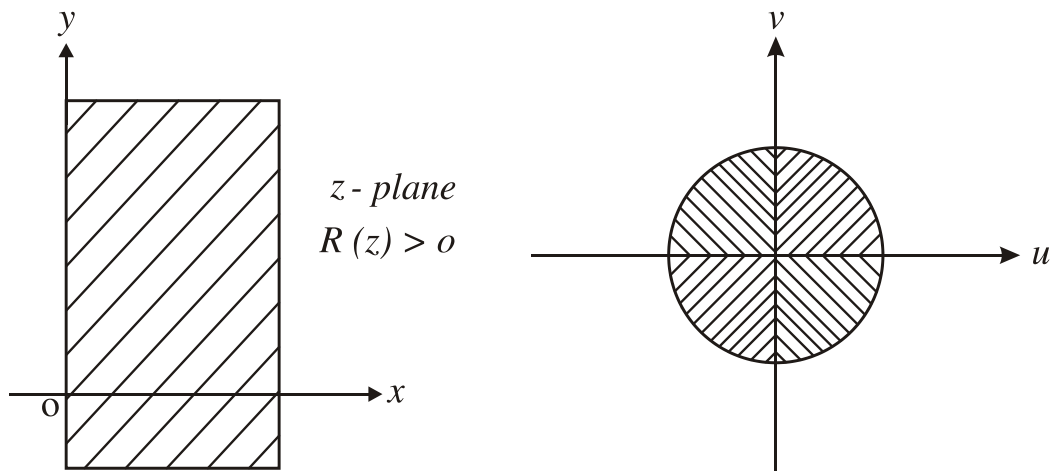


Figure - 2

Proof : Suppose the bilinear transformation

$$w = \frac{az + b}{cz + d} \dots\dots\dots (1)$$

transforms the half plane $R(z) \geq 0$ into the circle $|w| \leq 1$.

$$(1) \quad \text{is expressible as } w = \frac{a/c}{z + d/c} \frac{z + b/a}{1} \dots\dots\dots (2)$$

This $\Rightarrow c \neq 0$, otherwise the points at infinity in the two planes would correspond.

We have seen that the transformation (2) transforms a straight line of z -plane into a circle of w -plane and points symmetrical about the line transform into the inverse points w.r.t. $|w| = 1$.

Here the points z and $-\bar{z}$ symmetrical about the imaginary axis $x = R(z) = 0$ will correspond to $w = 0$ and $w = \infty$, the inverse points of the circle $|w| = 1$.

Thus we may write $-b/a = \alpha$, $-d/c = -\bar{\alpha}$ on putting above in eqⁿ (2), we get $w = 0$, $w = \infty$.

$$\text{Then (2) takes the form } w = \frac{a}{c} \left(\frac{z - \alpha}{z + \bar{\alpha}} \right) \dots\dots\dots (3)$$

The point $z = 0$ on the boundary of the half plane $R(z) \geq 0$ must correspond to a point on the boundary of $|w| = 1$ so that

$$1 = |w| = \left| \frac{a}{c} \right| \cdot \left| \frac{0 - \alpha}{0 + \bar{\alpha}} \right| = \left| \frac{a}{c} \right| \text{ so that } \frac{a}{c} = e^{i\lambda},$$

Where λ is real.

$$\text{Hence } w = e^{i\lambda} \left(\frac{z - \alpha}{z + \bar{\alpha}} \right) \dots\dots\dots (4)$$

evidently $z = \alpha$ gives $w = 0$. But $w = 0$ is an interior point of circle $|w| = 1$. Hence $z = \alpha$ must be a point of the right half plane i.e. $R(\alpha) > 0$. With this condition (4) is the required transformation.

Verification :

$$\begin{aligned}
 w \bar{w} - 1 &= \left(\frac{z - \alpha}{z + \bar{\alpha}} \right) e^{i\lambda} \left(\frac{\bar{z} - \bar{\alpha}}{\bar{z} - \alpha} \right) e^{-i\lambda} - 1 \\
 &= \frac{z\bar{z} + \alpha\bar{\alpha} - z\bar{\alpha} - \alpha\bar{z} - (z\bar{z} + \alpha\bar{\alpha} + z\alpha + \bar{z}\bar{\alpha})}{(z + \alpha)(\bar{z} + \alpha)} \\
 &= \frac{-z(\alpha + \bar{\alpha}) - \bar{z}(\alpha - \bar{\alpha})}{|z + \bar{\alpha}|^2} \\
 &= \frac{-(z + \bar{z})(\alpha + \bar{\alpha})}{|z + \bar{\alpha}|^2} = \frac{-2R(\alpha).2R(z)}{|z + \bar{\alpha}|^2} \\
 \text{or} \quad &= \frac{4R(\alpha).R(z)}{|z + \bar{\alpha}|^2} = 1 - |w|^2 \quad \dots\dots\dots (5)
 \end{aligned}$$

Also we have proved that $R(\alpha) > 0$ now from (5) it is clear that

(i) $R(z) = 0$ corresponds to $1 - |w|^2 = 0$ or $|w| = 1$

(ii) $R(z) > 0$ corresponds to $1 - |w|^2 > 0$ or $|w| < 1$

finally $R(z) > 0$ corresponds to $|w| \leq 0$, by virtue of transformation (4).

2.7.3 Find all the bilinear (Möbius) transformation which transforms the unit circle $|z| \leq 1$ into the unit circle $|w| \leq 1$.

or

Find the general homographic transformation which leaves the unit circle invariant.

Proof : Suppose the transformation

$$w = \frac{az + b}{cz + d} \quad \dots\dots\dots (1)$$

transforms the unit circle $|z| \leq 1$ into the circle $|w| \leq 1$,

$$w = \frac{a \left(z + \frac{b}{a} \right)}{c \left(z + \frac{d}{c} \right)} \quad \dots\dots\dots (2)$$

Clearly $c \neq 0$, otherwise the point at ∞ in the two plane will correspond. We have seen that the transformation (2) maps a circle of z -plane into a circle of w -plane and inverse points transform into inverse points.

$$\text{If } w = 0, \text{ then } c \left(z + \frac{d}{c} \right) = 0 \Rightarrow z = -\frac{d}{c}$$

$$\text{If } w = \infty, \text{ then } a \left(z + \frac{b}{a} \right) = 0 \Rightarrow z = -\frac{b}{a}$$

Thus $w = 0, \infty$, the inverse points of $|w| = 1$ are the transform of $z = -\frac{b}{a}$ and $z = -\frac{d}{c}$ respectively.

$$\text{Hence we may write } -\frac{b}{a} = \alpha, \quad -\frac{d}{c} = \frac{1}{\alpha}$$

Such that $|\alpha| < 1$

[For α , and $\frac{1}{\alpha}$ are the inverse points of circle $|z| = 1$]

$$\text{Hence from (2) } w = \frac{a}{c} \frac{z - \alpha}{z - \left(\frac{1}{\alpha} \right)}$$

$$\text{or } w = \frac{a\bar{\alpha} (z - \alpha)}{c (z\bar{\alpha} - 1)} \quad \dots\dots\dots (3)$$

The point $z = 1$ on the boundary of $|z| = 1$ must correspond to a point on the boundary of $|w| = 1$

$$\text{So that } 1 = |w| = \left| \frac{a\bar{\alpha}}{c} \right| \cdot \left| \frac{1-\alpha}{\bar{\alpha}-1} \right|$$

$$= \left| \frac{a\bar{\alpha}}{c} \right| \text{ or } \frac{a\bar{\alpha}}{c} = e^{i\lambda}$$

$$\text{Hence (3) becomes } w = e^{i\lambda} \left(\frac{z-\alpha}{\bar{\alpha}z-1} \right) \text{ with } |\alpha| < 1 \quad \dots\dots\dots(4)$$

This is the required transformation.

Verification

$$\begin{aligned} \omega\bar{\omega}-1 &= \left(\frac{z-\alpha}{\bar{\alpha}z-1} \right) e^{i\lambda} \left(\frac{\bar{z}-\bar{\alpha}}{\alpha\bar{z}-1} \right) e^{-i\lambda} - 1 \\ &= \frac{(z-\alpha)(\bar{z}-\bar{\alpha})}{(\bar{\alpha}z-1)(\alpha\bar{z}-1)} - 1 \\ &= \frac{z\bar{z} - \bar{\alpha}z - \alpha\bar{z} + \alpha\bar{\alpha} - (\bar{\alpha}\alpha z\bar{z} - \alpha\bar{z} - \bar{\alpha}z + 1)}{|\bar{\alpha}z-1|^2} \\ &= \frac{(z\bar{z}-1)(1-\alpha\bar{\alpha})}{|\bar{\alpha}z-1|^2} \end{aligned}$$

$$\text{or } |w|^2 - 1 = \frac{(|z|^2-1)(1-|\alpha|^2)}{|\bar{\alpha}z-1|^2} \quad \dots\dots\dots (5)$$

Also $|\alpha| < 1$ so that $1-|\alpha|^2 > 0$

Now from (5) it is clear that

(i) $|w| = 1$ corresponds to

$$\frac{(|z|^2-1)(1-|\alpha|^2)}{|\bar{\alpha}z-1|^2} = 0 \text{ or } |z|^2 - 1 \text{ or } |z| = 1$$

(ii) $|\omega| < 1$ or $|\omega|^2 - 1 < 0$ corresponds to

$$\frac{(|z|^2 - 1)(1 - |\alpha|^2)}{(\bar{\alpha}z - 1)^2} < 0 \text{ or } |z| < 1$$

finally, $|z| \leq 1$ is transformed to $|\omega| \leq 1$ by virtue to transformation (4)

2.7.4 The general linear transformation of the circle $|z| \leq \rho$ into the circle

$$|\omega| \leq e' \text{ is } \omega = ee' e^{i\lambda} \frac{(z - \alpha)}{\bar{\alpha}z - \rho^2}$$

Such that $|\alpha| < \rho$

Proof : Suppose the transformation

$$\omega = \frac{az + b}{cz + d} \dots\dots\dots (1) \text{ transforms } |z| \leq \rho$$

onto the circle $|\omega| \leq \rho'$

$$\text{from (1) } \omega = \frac{a/c}{z + d/c} \frac{z + b/a}{z + d/c} \dots\dots\dots (2)$$

Clearly $c \neq 0$, otherwise the point at infinity in the two planes would correspond.

We have seen that transformation (2) maps a circle of z -plane into a circle of ω -plane and inverse points transform into inverse points,

$$\text{If } \omega = 0 \text{ then (2) } z = -b/a$$

$$\text{If } \omega = \infty \text{ then (2) } z = -d/c$$

Thus $\omega = 0$, $\omega = \infty$, the inverse points of $|\omega| = e'$ are the transforms of

$z = -b/a$, $z = -d/c$ respectively. Hence we may write $-b/a = \alpha$, $-d/c = \rho^2/\alpha$ such that

$|\alpha| < \rho$.

[For α and $\frac{\rho^2}{\alpha}$ are the inverse point of circle $|z| = \rho$]

Hence, by (2),

$$\omega = \frac{a}{c} \frac{z - \alpha}{z - \rho^2/\bar{\alpha}} \text{ or } \omega = \frac{\bar{\alpha}a}{c} \left(\frac{z - \alpha}{\bar{\alpha}z - \rho^2} \right) \dots\dots\dots (3)$$

(3) Satisfies the condition $|z| \leq \rho$ and $|\omega| \leq \rho'$.

Hence for $|z| = \rho$, we must have $|\omega| = \rho'$ so that (3) becomes

$$\rho' = |\omega| = \left| \frac{\bar{\alpha}a}{c} \right| \left| \frac{z - \alpha}{\bar{\alpha}z - \rho^2} \right| = \left| \frac{\bar{\alpha}a}{c} \right| \left| \frac{1}{z} \right| \left| \frac{z - \alpha}{\bar{z} - \bar{\alpha}} \right| \text{ as } z\bar{z} = \rho^2.$$

But $|z - \alpha| = |\bar{z} - \bar{\alpha}|$

Hence $ee' = \left| \frac{\bar{\alpha}a}{c} \right|$ and $\frac{\bar{\alpha}a}{c} = ee'e^{i\lambda}$

Putting this in (3),

$$\omega = ee'e^{i\lambda} = \left(\frac{z - \alpha}{z\bar{\alpha} - \rho^2} \right) \text{ such that } |\alpha| < \rho \dots\dots\dots (4)$$

Where λ is real, this is the required transformation.

Verification

$$\begin{aligned} \omega \bar{\omega} - (e')^2 &= ee'e^{i\lambda} \left(\frac{z - \alpha}{z\bar{\alpha} - \rho^2} \right) ee'e^{-i\lambda} \left(\frac{\bar{z} - \bar{\alpha}}{\bar{z}\alpha - \rho^2} \right) ee' \\ &= \frac{e'^2}{|z\bar{\alpha} - \rho^2|^2} \left[\rho^2 (z - \alpha)(\bar{z} - \bar{\alpha}) - (z\bar{\alpha} - \rho^2)(\bar{z}\alpha - \rho^2) \right] \\ \Rightarrow |\omega|^2 - (\rho')^2 &= \rho'^2 \frac{(z\bar{z} - \rho^2)(\rho^2 - \alpha\bar{\alpha})}{|z\bar{\alpha} - \rho^2|^2} \end{aligned}$$

$$\text{or } |\omega|^2 - (\rho')^2 = \rho'^2 \frac{(|z|^2 - \rho^2)(\rho^2 - |\alpha|^2)}{|z\bar{\alpha} - \rho^2|^2} \dots\dots\dots (5)$$

Also we have proved that $|\alpha| < \rho$ or $\rho^2 - |\alpha|^2 > 0$

$$\begin{aligned} \text{from (5)} \quad |\omega| = \rho' &\Rightarrow |\omega|^2 - \rho'^2 = 0 \\ &\Rightarrow |z|^2 - \rho^2 = 0 \Rightarrow |z| = \rho \end{aligned}$$

$$\begin{aligned} \text{Again} \quad |w| < \rho' &\Rightarrow |w|^2 - \rho'^2 < 0 \Rightarrow |z|^2 - \rho^2 < 0 \\ &\Rightarrow \rho|z| < \rho \end{aligned}$$

Thus $|\omega| \leq \rho'$ corresponds to $|z| \leq \rho$

2.7.5 To show that the region $|z - \alpha| \leq R$ is mapped conformally on $|\omega| \leq 1$ by the transformation.

$$\omega = \frac{R(z - c)e^{i\lambda}}{R^2 - (z - a)(\bar{c} - \bar{a})}$$

where α is real and $z = c$ is the point which is transformed into the origin.

Proof : Suppose the transformation

$$\omega = \frac{Az + B}{Cz + D} \dots\dots\dots (1)$$

transforms $|z - a| \leq R$ onto $|\omega| \leq 1$ such that $z = c$ transformed into the origin.

$$\Rightarrow \omega = \frac{A}{C} \cdot \frac{z + B/A}{z + D/C} \dots\dots\dots (2)$$

Evidently $C \neq 0$, otherwise points at infinities in the two planes would correspond.

We have seen that the transformation (2) transforms a circle of z plane into a circle of w -plane and inverse points transforms into inverse-points.

If $w = 0$, then (2) $\Rightarrow z = -B/A$

If $w = \infty$, then (2) $\Rightarrow z = -D/C$

Thus $w = 0$, $w = \infty$, the inverse point of $|w| = 1$ are the transforms of $z = -B/A$, $z = -D/C$ respectively.

But it is given that $z = c$ corresponds to $w = 0$.

The inverse of $z = c$ relative to $|z - a| = R$ is given by

$$(z - a)(\bar{z} - \bar{a}) = R^2$$

$$\text{or } z = a + \left[\frac{R^2}{(\bar{c} - \bar{a})} \right]$$

Now, we may write $\frac{-B}{A} = c$, $\frac{-D}{C} = a + \frac{R^2}{\bar{c} - \bar{a}}$

$$\text{Now (2) is reduced to } w = \frac{A}{C} \left[\frac{z - c}{z - a - \left[\frac{R^2}{\bar{c} - \bar{a}} \right]} \right]$$

$$\text{or } w = \frac{A(\bar{c} - \bar{a})(z - c)}{C \left[(z - a)(\bar{c} - \bar{a}) - R^2 \right]} \dots\dots\dots (3)$$

$w\bar{w} = |w|^2 = 1$ corresponds to $(z - a)(\bar{z} - \bar{a}) = R^2$

Then (3) gives $1 = |w|$

$$= \left| \frac{A}{C} \right| \frac{|\bar{c} - \bar{a}| \cdot |z - c|}{|(z - a)(\bar{c} - \bar{a}) - (z - a)(\bar{z} - \bar{a})|} \quad \text{or} \quad \left| \frac{A}{C} \right| \frac{|\bar{c} - \bar{a}| \cdot |z - c|}{|z - a| |\bar{c} - \bar{z}|} = 1$$

But $|z - c| = |\bar{z} - \bar{c}|$ and $|z - a| = R$ Hence the

last gives, $\left| \frac{A(\bar{c} - \bar{a})}{c} \right| = R$ or $\frac{A(\bar{c} - \bar{a})}{c} = R e^{i\lambda}$ where λ is real.

$$\therefore \text{By (3), } \omega = \frac{R e^{i\lambda} (z - c)}{(z - a)(\bar{c} - \bar{a}) - R^2}$$

$$\text{or } \omega = \frac{R e^{i\lambda} (z - c)}{R^2 - (z - a)(\bar{c} - \bar{a})} \text{ which is the result.}$$

$$\text{For } -e^{i\lambda} = e^{i\pi} \cdot e^{i\lambda} = e^{i\alpha}$$

where $\alpha = \pi + \lambda$ is real.

Verification

Let $|z - a| < R$ so that

$$(z - a)(\bar{z} - \bar{a}) < R^2$$

$$|\omega| = \frac{R|z - c| \cdot |e^{i\alpha}|}{\left| R^2 - (z - a)(\bar{c} - \bar{a}) \right|} = \frac{R|z - c| \cdot 1}{\left| R^2 - (z - a)(\bar{c} - \bar{a}) \right|}$$

$$< \frac{R|z - c|}{\left| (z - a)(\bar{z} - \bar{a}) - (z - a)(\bar{c} - \bar{a}) \right|}$$

$$= \frac{R|z - c|}{|z - a||\bar{z} - \bar{a}|} = \frac{R}{|z - a|} < 1 \quad \text{or} \quad |\omega| < 1$$

finally, $|z - a| < R$ corresponds to $|\omega| < 1$

Thus the interiors correspond.

2.8 Conformal Mappings

Suppose the transformation $u = u(x, y)$

$$v = v(x, y)$$

maps the two curves C_1, C_2 [intersecting at the point $P(z_0)$] of z -plane on the curves $C'_1 C'_2$ [intersecting at $P'(w_0)$] of w plane

If the angle between C_1 and C_2 and at z_0 is equal to the angle between C'_1 and C'_2 at w_0 , then the transformation is called **isogonal**. If the sense of rotation as well as the magnitude of the angle is preserved, the transformation is called **Conformal**. In figure (A), the transformation is **conformal**.

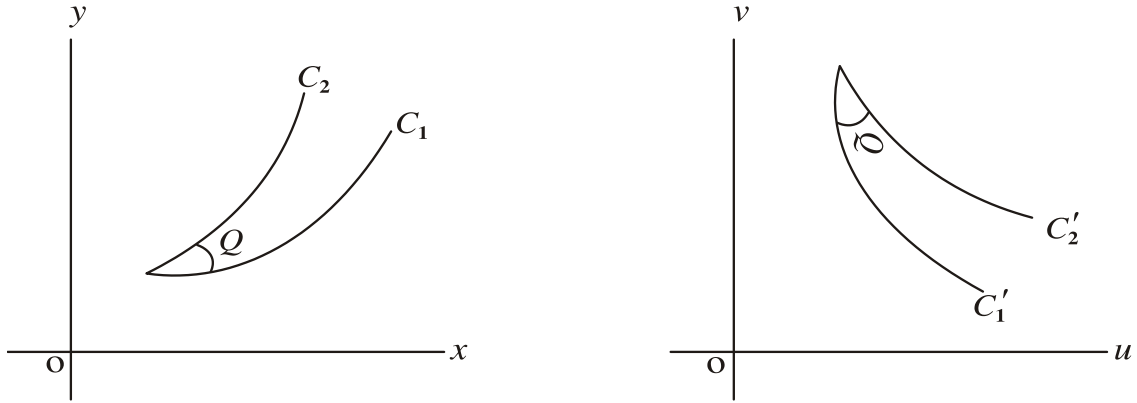


Figure (A)

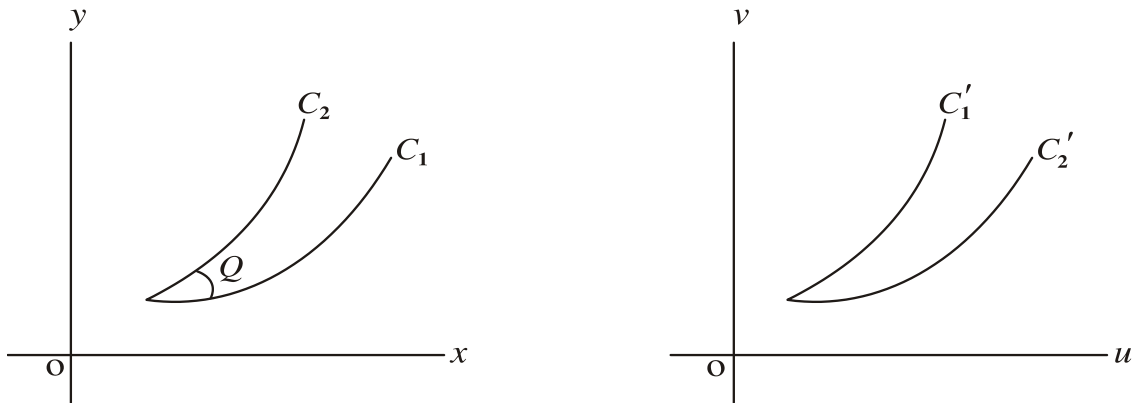


Figure (B)

While in figure (B) the transformation is **isogonal**. $\omega = \bar{z}$ is isogonal transformation.

2.9. The necessary and sufficient condition for the transformation $\omega = f(z)$ to be conformal is that $f(z)$ is analytic.

First suppose that $\omega = f(z)$ be an analytic function in a domain D on the z -plane and z_0 be an interior of D . Also C'_1 and C'_2 be the curves (intersecting at ω_0) in ω -plane. Corresponding to the two curves C_1 and C_2 (intersecting at z_0) of z -plane. Let ω_1 and ω_2 be points on C'_1 and C'_2 respectively corresponding to the

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points z_1, z_2 on c_1 and c_2 respectively such that Distance between z_1 and z_0 = distance between z_2 and $z_0 = r$ (suppose).

So we can write $z_1 - z_0 = re^{i\theta_1}$, $z_2 - z_0 = re^{i\theta_2}$ let the tangents at z_0 to the curves c_1^+ and c_2^+ make angles α_1, α_2 with the real axis so that $\theta_1 \rightarrow \alpha_1, \theta_2 \rightarrow \alpha_2$ as $r \rightarrow 0$.

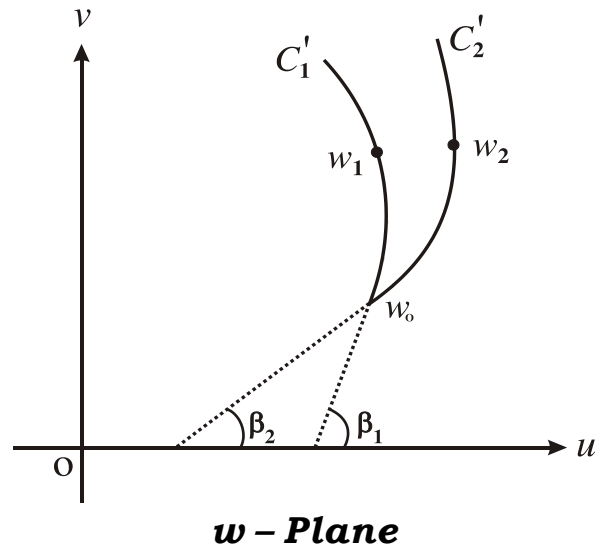
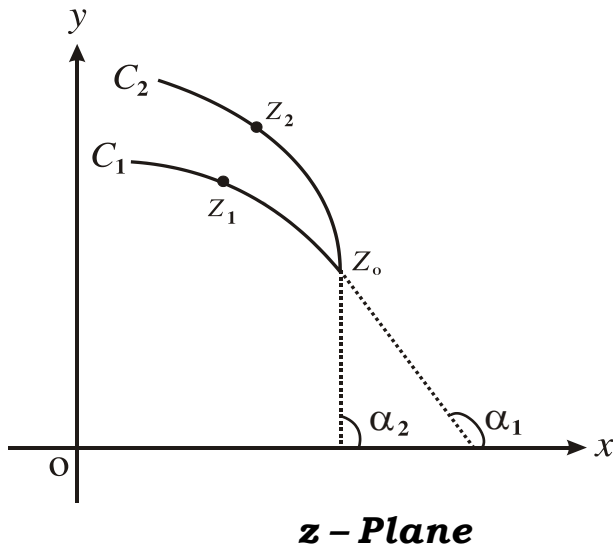
Also let the tangents at w_0 to the curves c_1' and c_2' make angle β_1 and β_2 with the real axis and let $w_1 - w_0 = \rho_1 e^{i\phi_1}$, $w_2 - w_0 = \rho_2 e^{i\phi_2}$ where $\phi_1 \rightarrow \beta_1$ as $\rho_1 \rightarrow 0$ and $\phi_2 \rightarrow \beta_2$ as $\rho_2 \rightarrow 0$.

$$\therefore f(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

$$= \lim_{z_1 \rightarrow z_0} \frac{w_1 - w_0}{z_1 - z_0} = \lim_{z_1 \rightarrow z_0} \frac{\rho_1 e^{i\phi_1}}{r e^{i\theta_1}}$$

$$\text{or, } f'(z_0) = \lim_{z_1 \rightarrow z_0} \frac{\rho_1}{r} e^{i(\phi_1 - \theta_1)}$$

By assumption $f'(z_0) \neq 0$. Write $f'(z_0) = R e^{i\lambda}$



$$\text{Then } R e^{i\lambda} = \lim_{z_1 \rightarrow z_0} \left[\frac{\rho_1}{r} e^{i(\phi_1 - \theta_1)} \right] = \beta_1 - \alpha_1$$

equating the modulus and argument,

$$R = \lim_{z_1 \rightarrow z_0} \left(\frac{\rho_1}{r} \right), \lambda = \lim (\phi_1 - \theta_1) = \beta_1 - \alpha_1 \text{ or, } \beta_1 = \lambda + \alpha_1$$

Similarly, we can show that $\beta_2 = \lambda + \alpha_2$

from which $\beta_1 - \beta_2 = \alpha_1 - \alpha_2$.

This proves that the angles between c_1' and c_2' at ω_0 is equal in magnitude and as well as in sign to the angle between the curves c_1 and c_2 . Therefore the transformation is conformal.

Again let us suppose that the mapping $\omega = f(z)$ is conformal then here we shall see that $f(z)$ is an analytic function of z .

Proof : Let us suppose $\omega = u(x, y) + iv(x, y) = f(z)$.

Also suppose $u = u(x, y)$, $v = v(x, y)$ are equations defining conformal transformation from z -plane to ω -plane. Let ds and $d\sigma$ be the line of elements in z -plane and ω -plane respectively so that $ds^2 = dx^2 + dy^2$, $d\sigma^2 = du^2 + dv^2$.

$$\partial u = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$\partial v = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$

Squaring and adding $d\sigma^2 = du^2 + dv^2$

$$= \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right] dx^2 + \left[\left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] dy^2 + 2 \left[\frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} \right] dx dy$$

Since transformation is conformal and hence the ratio $\partial\sigma^2 : \partial s^2$ is independent of direction. Comparing the last equation with $ds^2 = dx^2 + dy^2$

$$\frac{\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2}{1} = \frac{\left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2}{1}$$

$$= \frac{\frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \cdot \frac{\partial v}{\partial y}}{0}$$

$$\text{This} \Rightarrow \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 = \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \quad \dots\dots\dots (1)$$

$$\text{and} \quad \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \cdot \frac{\partial v}{\partial y} = 0 \quad \dots\dots\dots (2)$$

$$\text{from (2) } \frac{\frac{\partial u}{\partial x}}{\frac{\partial v}{\partial y}} = \frac{\partial v / \partial x}{-\partial u / \partial y} = \lambda \text{ (suppose)}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \lambda \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\lambda \frac{\partial u}{\partial y} \quad \dots\dots\dots (3)$$

Putting this in (1),

$$\lambda^2 \left(\frac{\partial v}{\partial y} \right)^2 + \lambda^2 \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2$$

$$\text{or, } (\lambda^2 - 1) \left[\left(\frac{\partial u}{\partial y} \right)^2 - \left(\frac{\partial v}{\partial y} \right)^2 \right] = 0$$

$$\Rightarrow \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda = \pm 1$$

Using this in (3),

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ where } \lambda = 1 \quad \dots\dots\dots (4)$$

$$\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \text{ where } \lambda = -1 \quad \dots\dots\dots (5)$$

The equation (4) are Cauchy-Riemann equations and hence $\omega = f(z)$ is an analytic function. The equation (5) are reduced to (4) by writing $-hv$ in place of v , i.e by taking as image figure obtained by the reflection in the real axis of ω -plane.

The mappings $\omega = z^n$ (n being a +ve integer) $\frac{d\omega}{dz} = n z^{n-1} = 0$ at $z = 0$ so that the above mapping is conformal at every point except at the point $z = 0$ where $\frac{d\omega}{dz} = 0$.

It is also to be noted as $z = \omega^{1/n}$ therefore for a given value of ω we will have n value of ω we will have n values of z showing therefore that the correspondence between z and ω is also not one-to-one but n to 1.

Writing $z = x + iy = ze^{i\theta}$ and $w = u + iv = \rho e^{i\phi}$, transformation can be written as $\rho e^{i\phi} = r^n e^{in\theta}$. Equating modulus and argument on both sides,

$$\text{we get } \rho = r^n \text{ and } \phi = n\theta \quad \dots (1)$$

It is now easy to verify the following facts about the mapping.

(A) From the relation $\rho = r^n$, we see that the concentric circles $r = \lambda$ in the z -plane for different values of λ correspond to the concentric circles $\rho = \lambda^n$ in the w -plane.

(B) The straight lines $\theta = \alpha$ through origin for varying values of α change over the straight line $\phi = n\alpha$ through the origin. Here the angle between the two straight lines through the origin is not preserved but is multiplied by n . Thus the conformal property does not hold at $z = 0$. It follows that a circular sector (wedge) with origin as vertex is transformed into a circular sector with origin as vertex but central angle becomes n times.

(C) The interior of the circular sector defined $0 < \arg z < \pi/n$ is mapped conformally on to $0 < \arg w < \pi$. Onto the entire upper half plane $I(w) > 0$.

Also the interior of the circular sector defined by

$$0 < \arg z < \frac{2\pi}{n}$$

is mapped conformally onto the entire w -plane cut along the entire positive real axis. The cut is introduced to avoid the origin where the conformal property does not hold. Such a cut is often called a slit and the plane is returned to as slit along positive real axis.

2.10. The mapping $w = z^n$, $w = z^2$ and inverse mapping $w = z^{1/2}$.

2.10.1 The transformation $w = z^2$.

Here this transformation will be discussed in detail.

$$\therefore w = z^2 \Rightarrow \frac{dw}{dz} = 2z = 0 \text{ at } z = 0$$

So that the conformal property does not hold at the origin. Here it is convenient to consider the cartesian form $z = x + iy$, $w = u + iv$.

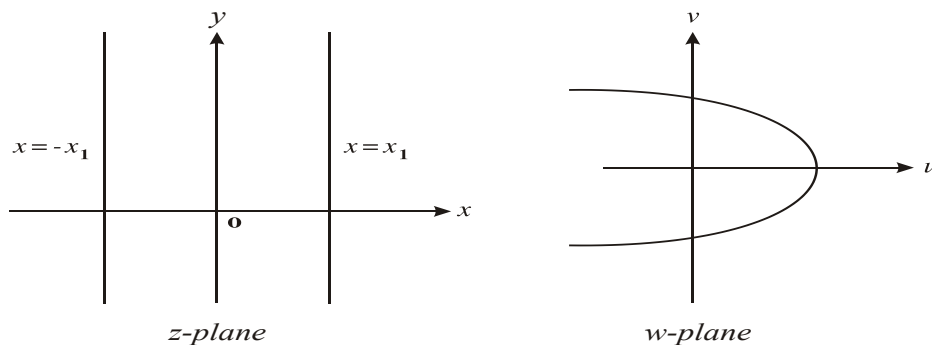
Substituting these values in the given transformation we have

$$u + iv = (x + iy)^2 = x^2 - y^2 + 2i xy.$$

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Hence equating the real and imaginary parts, we get $u = x^2 - y^2$ and $v = 2xy$.

(i) We first consider the images of lines $x = x_1$, $x = x_2$, the line $x = x_1 > 0$ is transformed into the curve $u = x_1^2 - y^2$, $v = 2x_1y$.



Where y is regarded as a parameter. Eliminating y , we get

$$u = x_1^2 - \frac{v^2}{4x_1^2}$$

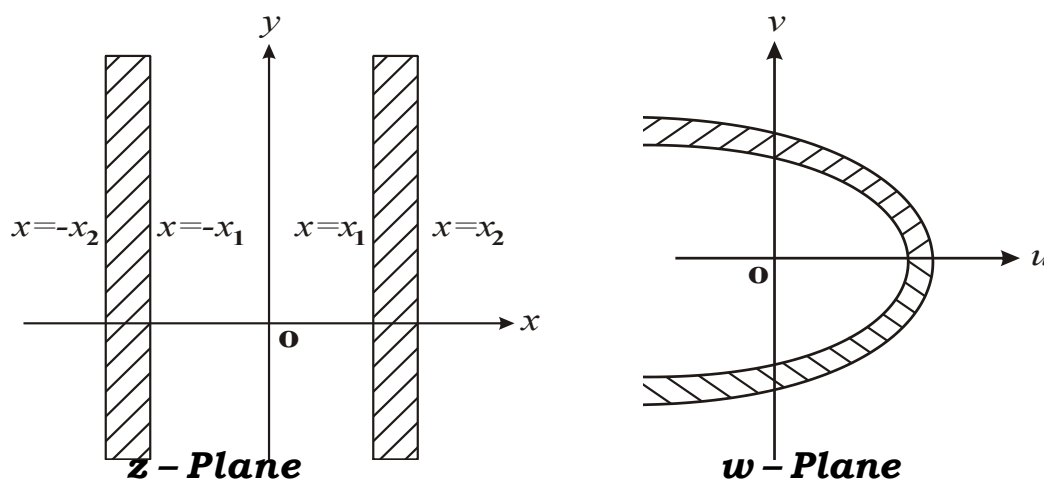
$$\text{or, } v^2 = -4x_1^2 (u - x_1^2)$$

Which is a parabola in the w -plane with its vertex at $(x_1^2, 0)$, focus at the origin and symmetrical about the real axis. Observe that the line $x = x_1$ is also transformed into the same parabola.

It is also evident that the two parts of the line $x = \pm x_1$ lying above and below the real axis separately correspond to the two parts of the parabola lying above and below the real axis in the w -plane.

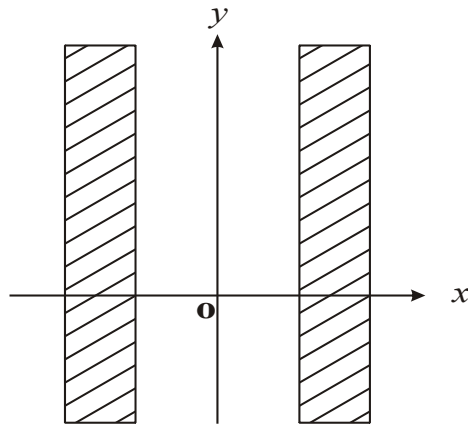
(ii) Both the infinite strips defined by $x_1 < x < x_2$ and $x_2 < x < -x_1$ are conformally mapped on the domain in the w -plane included between the parabolas.

$$u^2 = -4x_1^2 (u - x_1^2), \quad v^2 = -4x_2^2 (u - x_2^2).$$

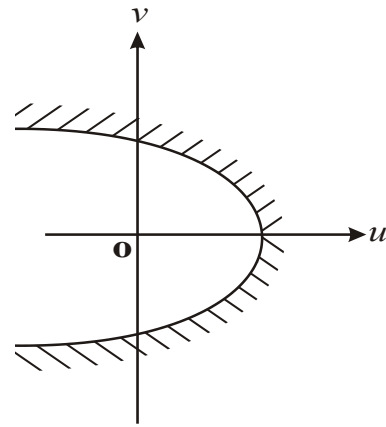


If we make x_2 tend to infinity, we see that the half plane to the right of $x = x_1$ i.e. the domain defined by $R(z) > x_1$ is mapped conformally on the exterior of the parabola $v^2 = -4x_1^2 (u - x_1^2)$ not containing the origin. The half plane defined by $x < x_1$ i.e., $R(z) < x_1$ is also mapped on the same exterior as shown on the previous page.

It is thus shown that the transformation $w = z^2$ maps the half plane $R(z) > x_1$ on to the exterior of the parabola $v^2 = -4x_1^2 (u - x_1^2)$ where $x_1 > 0$.

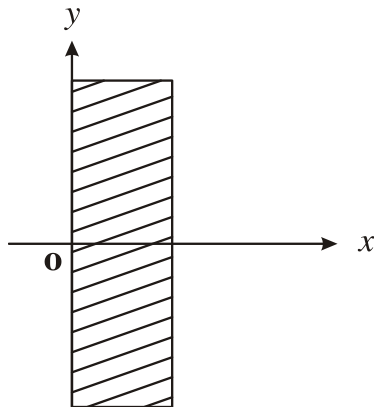


z - Plane

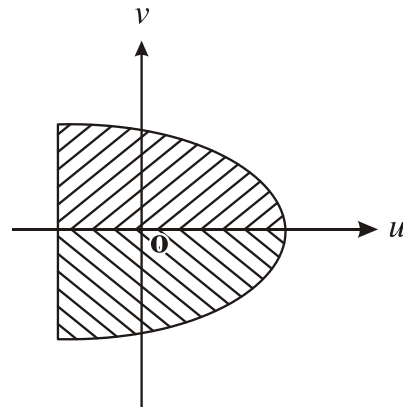


w - Plane

(iii) We now study the interior of this parabola. If we consider $x = x_1$ and make x_1 tend to zero. We easily see the domain $0 < x < x_1$ is conformally mapped on to the whole interior of the parabola $u^2 = -4x_1^2 (u - x_1^2)$ with a cut along the negative real units from $-\infty$ to 0 as shown below.



z - Plane



w - Plane

Combining the discussions given in (II) and (III) we can say that the half plane defined by $R(z) > 0$ is mapped onto the whole w plane cut along the negative real axis from 0 to $-\infty$. The half plane $R(z) < 0$ is also mapped onto the same whole cut plane.

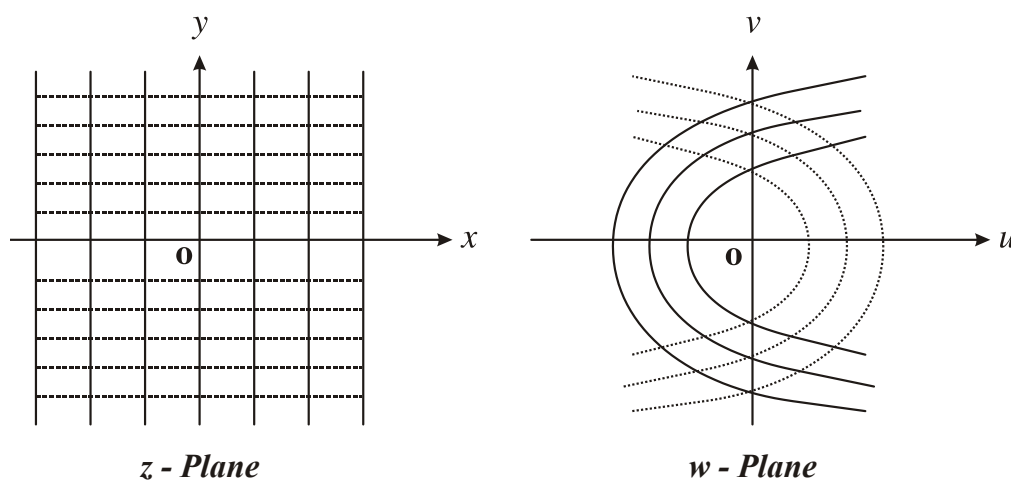
Bilinear Transformation

(iv) We leave it to the reader to verify that the image of the line $y = y_1$ is the parabola $v^2 = 4y_1^2 (u + y_1^2)$. Whose vertex is at $(-y_1^2, 0)$ and whose focus at the origin. Details similar to those in (i), (ii) and (iii) can be easily disussed. It will be seen as a final result that the half plane $f(z) > 0$ is mapped on the whole ω -plane cut along the positive real axis from 0 to ∞ . The half plane $f(z) < 0$ is also mapped on the same cut plane.

(v) The two families of parabolas.

$$v^2 = -4x_1^2 (u - x_1^2) \text{ and } v^2 = 4y_1^2 (u + y_1^2).$$

Corresponding to two orthogonal families of straight line $x = \pm x_1$ and $y = \pm y_1$



for varying values of (x_1, y_1) themselves from an orthogonal net, as is well known from analytic geometry. The families of straight lines and parabolas are shown in the above figures.

2.10.2 Transformation $\omega = \sqrt{z}$

$$\omega = \sqrt{z} \Rightarrow \omega^2 = z \Rightarrow (u + iv)^2 = x + iy$$

$$\Rightarrow u^2 - v^2 + iuv = x + iy$$

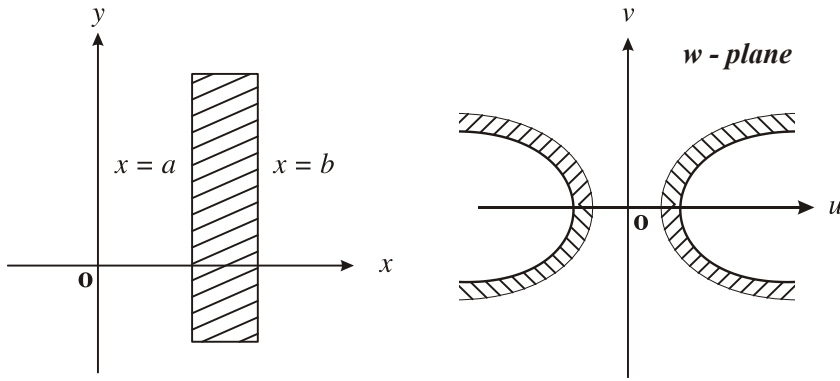
$$\Rightarrow u^2 - v^2 = x, \quad \dots (1)$$

$$2uv = y \quad \dots (2)$$

Case I : When $x = \text{constant}$.

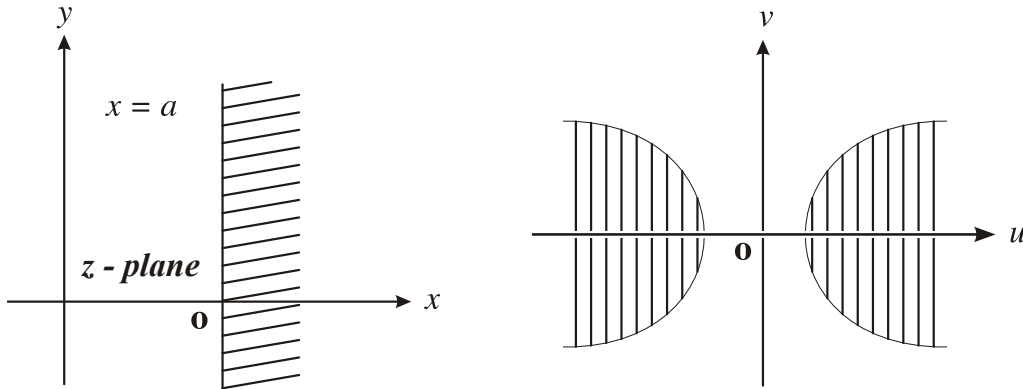
(a) From (1) it is clear that line $x = a$, where $a > 0$ in the z -plane corresponds to the hyperbola $u^2 - v^2 = a$ in ω -plane.

(b) The slit in z -plane between the lines $x = a, x = b$ where $a, b > 0$ and $b > a$ is mapped upon the area included between the hyperbolas $u^2 - v^2 = a, u^2 - v^2 = b$.



We have $u^2 - v^2 = x$, Hence $x = a$, $x = b$ gives $u^2 - v^2 = a$ and $u^2 - v^2 = b$. The lines $x = \lambda$ in z -plane as λ takes values continuously from a to b . This shaded area of z -plane is mapped upon the shaded area between the hyperbola in w -plane which is developed by the hyperbola $u^2 - v^2 = \lambda$ as λ varies from a to b .

(c) To prove that the domain in the z -plane to the right of the line $x = a$ is mapped upon the interior of the hyperbola in w -plane.



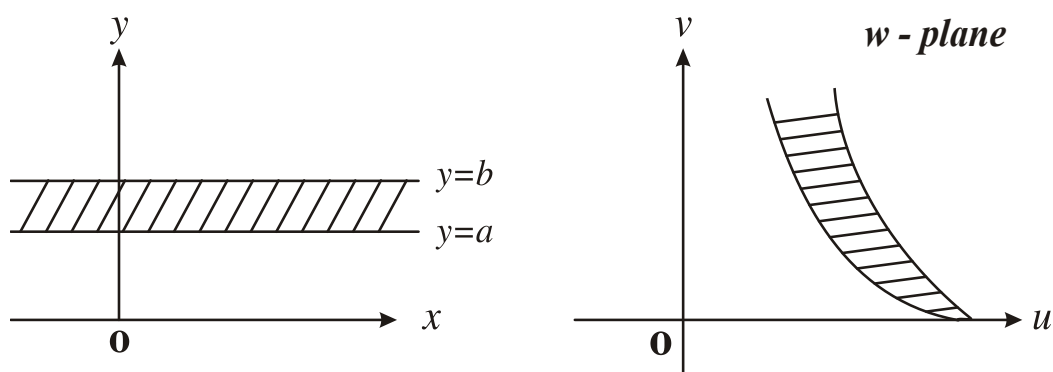
We have seen that the lines $x = \lambda$ in z -plane corresponds to the hyperbola $u^2 - v^2 = \lambda$ in w plane. Suppose λ takes the values continuously from a to ∞ . Then the area in z -plane developed by the line $x = \lambda$ will be the entire of z -plane lying to the right of the line $x = a$. This area will be mapped upon the area interior to the hyperbola $u^2 - v^2 = a$ in w -plane which is developed by $u^2 - v^2 = \lambda$.

Case II : When y -constant.

(a) The line $y = a$ corresponds to the rectangular hyperbola $2uv = a$ for $2uv = y$, $y = a \Rightarrow 2uv = a$.

(b) The slit in z -plane between the lines $y = a$ and $y = b$ where $a, b > 0$ and $b > a$ is mapped upon the area included between the rectangular hyperbolas $2uv = a$ and $2uv = b$

Bilinear Transformation



complete the proof. The proof is similar to (b) part of case I

2.11. Summary

In this unit, we have discussed.

- (i) Bilinear Transformation.
- (ii) Resultant or Product of two bilinear transformation.
- (iii) Cross Ratio.
- (iv) Preservation of family of straight lines and circles.
- (v) Fixed points and normal form of a bilinear Transformation.
- (vi) Some special bilinear Transformation.
- (vii) Conformal Mapping.
- (viii) The mapping $w = z^n$, $w = z^2$ and the inverse mapping $w = z^{1/2}$.

2.12. Solved examples on conformal mappings.

Example 1 : Find the fixed point and the normal form of the following bilinear transformation and classify their nature.

$$(a) w = \frac{3iz + 1}{z + i} \quad (b) w = \frac{3z - 4}{z - 1}$$

Solution : (a) Putting $w = z$, then fixed points are given by

$$z = \frac{3iz + 1}{z + i} \Rightarrow z^2 + iz - 3iz - 1 = 0$$

$$\Rightarrow z^2 - 2iz - 1 = 0 \Rightarrow z^2 - 2iz + i^2 = 0$$

$$\Rightarrow (z - i)^2 = 0$$

$$\text{or, } z = i, i.$$

Hence there is only one distinct fixed points namely i .

$$\begin{aligned}\text{Now } \omega - i &= \frac{3iz + 1}{z + i} - i = \frac{3iz + 1 - iz - i^2}{z + i} \\ &= \frac{2iz + 1 + 1}{z + i} = \frac{2iz + 2}{z + i} \\ \Rightarrow \frac{1}{\omega - i} &= \frac{z + i}{2iz + 2} = \frac{z + i}{2i(z - i)} = \frac{1}{2i} \left[\frac{z - i + 2i}{z - i} \right] \\ &= \frac{1}{2i} \left[1 + \frac{2i}{z - i} \right] = \frac{1}{2i} + \frac{1}{z - i}\end{aligned}$$

or, $\frac{1}{\omega - i} = \frac{1}{z - i} - \frac{i}{2}$ which is the normal form. The transformation is parabola.

Solution: (b) $\omega = \frac{3z - 4}{z - 1}$

Putting $\omega = z$, $z = \frac{3z - 4}{z - 1}$

$$\Rightarrow z(z - 1) = 3z - 4 \Rightarrow z^2 - z - 3z + 4 = 0$$

$$\Rightarrow (z - 2)^2 = 0.$$

Thus $z = 2$ is the only fixed point so the transformation is parabolic. For

normal form $\omega - 2 = \frac{3z - 4}{z - 1} - 2 = \frac{z - 2}{z - 1}$

$$\therefore \frac{1}{\omega - 2} = \frac{z - 1}{z - 2} = \frac{z - 2 + 1}{z - 2} = \frac{1}{z - 2} + 1$$

Above is the required normal form.

Example 2. Find the bilinear transformation which maps the points $z_1 = 2$, $z_2 = i$ and $z_3 = -2$ into the points $\omega_1 = 1$, $\omega_2 = i$ and $\omega_3 = -1$.

Solution : The required transformation is given by

$$\frac{(\omega - \omega_1)(\omega_2 - \omega_3)}{(\omega_1 - \omega_2)(\omega_3 - \omega)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} \quad \dots (1)$$

Substituting the given values as above (1) becomes

Bilinear Transformation

$$\frac{(\omega - 1)(i + 1)}{(1 - i)(-1 - \omega)} = \frac{(z - 2)(i + z)}{(z - 0)(-2 - z)}$$

$$\Rightarrow -\frac{(\omega - 1)(1 + i)(1 + i)}{(\omega + 1)(1 - i)(1 + i)} = -\frac{(z - 2)(2 + i)(2 + i)}{(z + 2)(2 - i)(2 + i)}$$

$$\Rightarrow \frac{(\omega - 1)(1 + i)^2}{(\omega + 1)[1 - i^2]} = \frac{(z - 2)(2 + i)^2}{(z + 2)(2^2 - i^2)}$$

$$\Rightarrow \frac{(\omega - 1).2i}{(\omega + 1).2} = \frac{(z - 2)(3 + 4i)}{(z + 2).5}$$

$$\Rightarrow \frac{\omega - 1}{\omega + 1} = \frac{(z - 2)(3 + 4i)}{(z + 2)5i}$$

$$\Rightarrow \frac{(\omega - 1)}{(\omega + 1)} = \frac{(z - 2)i(4 - 3i)}{5(z + 2)i}$$

$$\Rightarrow \frac{\omega - 1}{\omega + 1} = \frac{(z - 2)(4 - 3i)}{5(z + 2)}$$

Using componendo & dividendo, we get

$$\frac{\omega - 1 + (\omega + 1)}{(\omega + 1) - (\omega - 1)} = \frac{(z - 2)(4 - 3i) + 5(z + 2)}{5(z + 2) - (z - 2)(4 - 3i)}$$

$$\Rightarrow \frac{2\omega}{2} = \frac{4z - 8 - 3iz + 6i + 5z + 10}{5z + 10 - 4z + 8 + 3iz - 6i}$$

$$\Rightarrow \omega = \frac{9z - 3iz + 2 + 6i}{z + 3iz + 18 - 6i} = \frac{3(3 - i)z + 2(1 + 3i)}{z(1 + 3i) + 6(3 - i)}$$

$$= \frac{3z + 2 \frac{(1 + 3i)}{3 - i}}{\left(\frac{1 + 3i}{3 - i}\right)z + 6} = \frac{3z + 2i}{iz + 6}$$

which is required transformations.

Example 3. Find the bilinear transformation which carries 0, i , $-i$ and 1, -1 , 0.

Solution : Consider z_1, z_2, z_3 as $0, i, -i$ and $\omega_1, \omega_2, \omega_3$ as $1, -1, 0$. By cross ratio we have

$$\begin{aligned} \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} &= \frac{(\omega - \omega_1)(\omega_2 - \omega_3)}{(\omega_1 - \omega_2)(\omega_3 - \omega)} \\ \Rightarrow \frac{(z - 0)(i + i)}{(0 - i)(-i - z)} &= \frac{(\omega - 1)(-1 - 0)}{(1 + 1)(0 - \omega)} \\ \Rightarrow \frac{2iz}{i(i + z)} &= \frac{-(\omega - 1)}{-2\omega} \\ \Rightarrow 4 \omega z &= (\omega - 1)(z + i) \\ \Rightarrow 4 \omega z &= (\omega - 1)z + (\omega + 1)i \\ \Rightarrow 4 \omega z - \omega z + z &= + \omega i - i \\ \Rightarrow \omega (4z - z - i) &= -z - i \\ \Rightarrow \omega (3z - i) &= -z - i \\ \Rightarrow \omega &= \frac{-(z + i)}{-(-3z + i)} = \frac{z + i}{-3z + i} \quad \dots (1) \end{aligned}$$

Which is of the form $\omega = f(z) = \frac{az + b}{cz + d}$, at $ad - bc \neq 0$

The relation (1) is called bilinear transformation.

Example 4. Show that the transformation $\omega = \frac{2z + 3}{z - 4}$ transforms the circle $x^2 + y^2 - 4x = 0$ into the straight line $4u + 3 = 0$ and explain the curve obtained is not a circle.

Solution : We have $\omega = \frac{2z + 3}{z - 4}$ or $(z - 4)\omega = 2z + 3$

$$\Rightarrow \omega z - 4\omega = 2z + 3 \Rightarrow z(\omega - 2) = 3 + 4\omega$$

$$\Rightarrow z = \frac{3 + 4\omega}{\omega - 2}$$

The equation of the circle in z -plane is

$$x^2 + y^2 - 4x = 0$$

$$\Rightarrow z \bar{z} - 2(z + \bar{z}) = 0$$

Bilinear Transformation

By virtue of the given transformation this is transformed to

$$\frac{(3+4\omega)(3+4\bar{\omega})}{(\omega-2)(\bar{\omega}-2)} - 2 \left[\frac{3+4\omega}{\omega-2} + \frac{3+4\bar{\omega}}{\bar{\omega}-2} \right] = 0$$

$$\text{or, } 9 + 16\omega\bar{\omega} + 12(\omega + \bar{\omega}) - 2[(3+4\omega)(\bar{\omega}-2) + (3+4\bar{\omega})(\omega-2)] = 0$$

$$\text{or, } 9 + 16\omega\bar{\omega} + 12(\omega + \bar{\omega}) - 2[(6 + 4\omega\bar{\omega} - 3\bar{\omega} - 8\omega)(4\omega\bar{\omega} + 3\omega - 6 - 8\bar{\omega})] = 0$$

$$\text{or, } 22(\omega + \bar{\omega}) + 33 = 0$$

$$\text{or, } 2(\omega + \bar{\omega}) + 3 = 0$$

or, $4u + 3 = 0$ which is the eqn. of a straight line in ω -plane.

Hence curve thus obtained is not a circle.

Example 5. Find the image of circle $|z - 2| = 2$ under the Mobius transformation $\omega = \frac{z}{z+1}$.

$$\textbf{Solution : } \omega = \frac{z}{z+1} \Rightarrow \omega(z+1) = z \Rightarrow z(\omega-1) = -\omega$$

$$\Rightarrow z = \frac{\omega}{1-\omega} \quad \dots (1)$$

$$\text{Now } |z - 2| = 2 \text{ gives } (z-2)(\bar{z}-2) = z^2$$

$$\text{By (1), } z - 2 = \frac{\omega}{1-\omega} - 2 = \frac{\omega - 2 - 2\omega}{1-\omega} = \frac{3\omega - 2}{1-\omega}$$

$$\bar{z} - 2 = \frac{\bar{\omega}}{1-\bar{\omega}} - 2 = \frac{3\bar{\omega} - 2}{1-\bar{\omega}}$$

Putting these values in (2)

$$\left(\frac{3\omega - 2}{1-\omega} \right) \left(\frac{3\bar{\omega} - 2}{1-\bar{\omega}} \right) = 4$$

$$\text{or, } 9\omega\bar{\omega} + 4 - 6(\omega + \bar{\omega}) = 4[1 + \omega\bar{\omega} - (\omega + \bar{\omega})]$$

$$\text{or, } 5\omega\bar{\omega} - 2(\omega + \bar{\omega}) = 0$$

$$\text{or, } 5(u^2 + v^2) - 2.2u = 0 \text{ where } \omega = u + iv$$

$$\text{or, } u^2 + v^2 - \frac{4}{5}u = 0$$

Hence it is a circle with centre $\left(\frac{2}{5}, 0\right)$ and radius $= (g^2 + f^2 - c)^{1/2} = \frac{2}{5}$.

So Image is a circle with centre $\left(\frac{2}{5}, 0\right)$ and radius $= \frac{2}{5}$ in ω -plane.

Example 6. Discuss the application of transformation $\omega = \frac{iz + 1}{z + 1}$ to the areas in the z -plane which are respectively inside and outside the unit circle with its centre at the origin.

Solution : The given transformation is

$$\omega = \frac{iz + 1}{z + i} = \frac{i(x + iy) + 1}{x + iy + i} = \frac{ix - y + 1}{x + i(1 + y)}$$

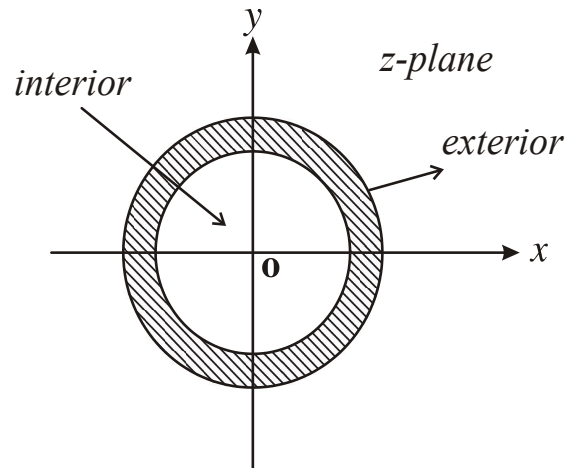
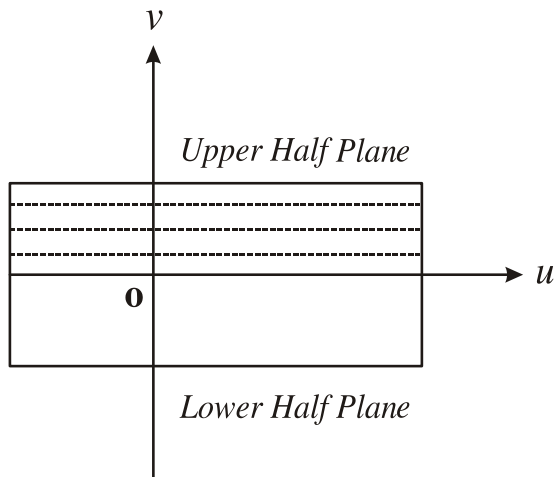
$$\Rightarrow u + iv = \left[\frac{(1 - y) + ix}{x + i(1 + y)} \right] \left[\frac{x - i(1 + y)}{x - i(1 + y)} \right]$$

$$= \frac{-i(1 - y^2) + ix^2 + x(1 - y) + x(1 + y)}{x^2 + (1 + y)^2}$$

Equating imaginary parts both sides only,

$$v = \frac{x^2 - (1 - y^2)}{x^2 + (1 + y)^2} = \frac{x^2 + y^2 - 1}{x^2 + (1 + y)^2} = \frac{|z|^2 - 1}{x^2 + (1 + y)^2}$$

$$\Rightarrow v = \frac{|z|^2 - 1}{x^2 + (1 + y)^2} \quad \dots (1)$$



Bilinear Transformation

The denominator of this fraction is essentially positive. From (1) it is clear that

(i) $v = 0$ (i.e., real axis of ω -plane) corresponds to $|z|^2 - 1 = 0$ or $|z| = 1$ (boundary of the circle $|z| = 1$).

(ii) $v > 0$ (upper half of ω -plane) corresponds to $|z|^2 - 1 > 0$ or $|z| > 1$ (exterior of the circle $|z| = 1$).

(iii) $v < 0$ (lower half of ω -plane) corresponds to $|z| < 1$ (interior of the circle $|z| = 1$).

This concludes the problem.

Example 7. Show that the transformation $\omega = \frac{5-4z}{4z-2}$ transform the circle $|z| = 1$ into a circle of radius unity in ω -plane and find the centre of the circle.

Solution : We have $\omega = \frac{5-4z}{4z-2}$ from which $4z\omega - 2\omega = 5 - 4z$

$$\text{or, } z = \frac{2\omega + 5}{4\omega + 4}$$

$\Rightarrow |z| = z\bar{z} = 1$ corresponds to

$$\left[\frac{2\omega + 5}{4(\omega + 1)} \right] \left[\frac{2\bar{\omega} + 5}{4(\bar{\omega} + 1)} \right] = 1$$

$$\text{or, } 4\omega\bar{\omega} + 25 + 10(\omega + \bar{\omega}) = 16(\omega\bar{\omega} + \omega + \bar{\omega} + 1) \dots (1)$$

But $\omega = u + iv$, $\bar{\omega} = u - iv$,

$$\omega\bar{\omega} = u^2 + v^2 \text{ and } \omega + \bar{\omega} = 2u.$$

Hence (1) becomes $4(u^2 + v^2) + 25 + 10 \cdot 2u$

$$= 16[u^2 + v^2 + 2u + 1]$$

$$\Rightarrow 4u^2 + 4v^2 + 25 + 20u - 16u^2 - 16v^2 - 32u - 16 = 0$$

$$\Rightarrow -12u^2 - 12v^2 - 12u + 9 = 0$$

$$\Rightarrow 12[u^2 + v^2 + u - 9/12] = 0$$

$$\Rightarrow u^2 + v^2 + u - 3/4 = 0 \quad \dots (2)$$

Comparing this with $x^2 + y^2 + 2gx + 2fy + c = 0$ we get centre as

$$(-g, -f) = \left(-\frac{1}{2}, 0 \right) \text{ and radius } = (g^2 + f^2 - c)^{1/2}, \text{ i.e., radius } = \left(\frac{1}{4} + 0 + \frac{3}{4} \right)^{1/2} = 1.$$

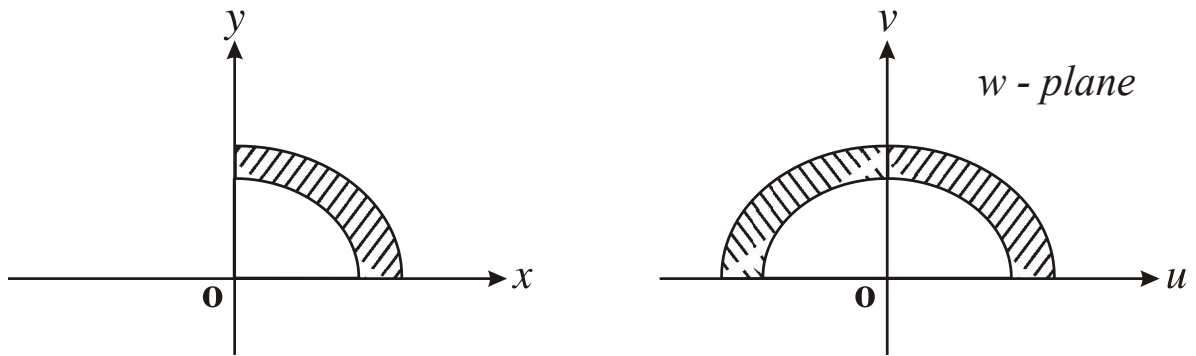
Hence (2) represents circle with its centre as $\left(-\frac{1}{2}, 0\right)$ and of radius one in ω -plane.

Problem on special Transformation.

Example 8. Discuss the application of the transformation $\omega = z^2$ to the area in the first quadrant of z -plane bounded by the axes and the circle $|z| = a$, $|z| = b$, ($a > b > 0$).

Is the transformation conformal ?

Solution :



The given transformation is $\omega = z^2$.

Taking $\omega = Re^{i\phi}$, $z = re^{i\theta}$ we get

$$Re^{i\phi} = r^2 e^{i2\theta}$$

$$\Rightarrow R = r^2, \phi = 2\theta$$

In view of this, $r = a \Rightarrow R = a^2$ and $0 \leq \theta \leq \frac{\pi}{2} \Rightarrow 0 \leq \phi \leq \pi$

Now the quadrants $|z| = a$, $0 \leq \theta \leq \frac{\pi}{2}$

and $|z| = b$, $0 \leq \theta \leq \frac{\pi}{2}$ in the z -plane,

are transformed into the semicircles $|\omega| = a^2$, $0 \leq \phi \leq \pi$ and $|\omega| = b^2$, $0 \leq \phi \leq \pi$ respectively of ω -plane.

From this it follows that the area/included in the first quadrant of z -plane is transformed into the area included between the semi-circles of ω -plane as shown in the figure.

The transformation $\omega = z^2$ is conformal.

$$\Rightarrow \text{For } f'(z) = \frac{d\omega}{dz} = 2z \neq 0 \text{ for any } z \text{ in the given area.}$$

Example 9. To show that the transformation

$$\omega = \tan^2 \left(\frac{\pi}{4} \sqrt{z} \right)$$

transforms the interior of unit's circle $|\omega| = 1$ into the interior of the parabola

$$r = \sec^2 \frac{\theta}{2}.$$

Solution : The transformation is $\omega = \tan^2 t$ where $t = \frac{\pi}{4} \sqrt{z}$

$$\omega = \frac{1 - \cos 2t}{1 + \cos 2t} \quad \dots (1)$$

$$2t = \frac{2\pi}{4} \sqrt{z} = \frac{\pi}{2} = \frac{\pi}{2} r^{\frac{1}{2}} e^{i\frac{\theta}{2}} = a + ib, \text{ say}$$

$$\text{Then } a = \frac{\pi}{2} \sqrt{r} \cos \frac{\theta}{2}$$

$$b = \frac{\pi}{2} \sqrt{r} \sin \frac{\theta}{2}$$

$$\cos 2t = \cos (a + ib) = \cos a \cosh b - i \sin a \sinh b = \cos a \cosh b - i \sin a \sinh b.$$

Now (1) becomes

$$\omega = \frac{1 - \cos a \cosh b + i \sin a \sinh b}{1 + \cos a \cosh b - i \sin a \sinh b}$$

$$\therefore |\omega| \leq 1 \Rightarrow \frac{(1 - \cos a \cosh b)^2 + \sin^2 a \sinh^2 b}{(1 + \cos a \cosh b)^2 + \sin^2 a \sinh^2 b} \leq 1$$

$$\text{or, } (1 - \cos a \cosh b)^2 + \sin^2 a \sinh^2 b \leq (1 + \cos a \cosh b)^2 + \sin^2 a \sinh^2 b$$

$$\text{or, } \cos a \cosh b \geq 0, \text{ but } \cosh b = \frac{e^b + e^{-b}}{2} > 0$$

$$\text{Hence, } \cos a \leq 0 \quad \text{or, } a \leq \frac{\pi}{2}$$

$$\text{or, } \frac{\pi}{2} \sqrt{r} \cos \frac{\theta}{2} \leq \frac{\pi}{2}$$

$$\text{or, } r \cos^2 \frac{\theta}{2} = 1 \text{ or } r \leq \sec^2 \frac{\theta}{2} \text{ or } r (1 + \cos \theta) \leq 2$$

$$\text{or, } \frac{2}{r} \geq 1 + \cos \theta \quad \dots\dots\dots (2)$$

Which represents interior and boundary of the parabola

$$r = \sec^2 \frac{\theta}{2} \text{ or, } \frac{2}{r} = 1 + \cos \theta.$$

This concludes the problem.

Example 10. If $\omega = 2z + z^2$, prove that the circle $|z| = 1$ correspond to a cardioid in ω -plane.

Solution : The given transformation is expressible as $\omega + 1 = (1 + z)^2$.

By writing $\omega + 1 = Re^{i\phi}$, i.e., by taking the pole on ω -plane at $\omega = -1$, we get

$$Re^{i\phi} = (1 + z)^2 \quad \dots\dots (1)$$

Any point on the circle $|z| = 1$ is expressible as $z = e^{i\theta}$, By the transformation (1), this is transformed to the curve $Re^{i\phi} = (1 + e^{i\theta})^2$

$$\begin{aligned} &= \left[e^{i\theta/2} \left(e^{-i\frac{\theta}{2}} + e^{i\theta/2} \right) \right]^2 \\ &= 4 \cos^2 \frac{\theta}{2} \cdot e^{i\theta} = 2 (1 + \cos \theta) e^{i\theta} \end{aligned}$$

$$\text{or, } Re^{i\phi} = 2 (1 + \cos \theta) e^{i\theta}$$

Equating the modulus and argument

$$R = 2 (1 + \cos \theta), \phi = \theta$$

$$\Rightarrow R = 2 (1 + \cos \phi)$$

Thus the unit circle $z = e^{i\theta}$ in z -plane corresponds to the cardioid $R = 2 (1 + \cos \phi)$ in ω -plane.

Example 11. Show that the transformation $\omega (z + i)^2 = 1$ maps interior of the circle $|z| = 1$ in the z -plane on the exterior of the parabola in ω -plane.

Solution : Suppose $z = re^{i\theta}$, $\omega = Re^{i\phi}$.

Now the given transformation is expressible as

$$z + i = \frac{1}{\sqrt{\omega}}$$

Bilinear Transformation

$$\text{or, } re^{j\theta} + i = \frac{1}{\sqrt{R}} e^{-i\phi/2}$$

Equating the real and imaginary parts

$$r \cos \theta = \frac{1}{\sqrt{R}} \cos \frac{\phi}{2}$$

$$r \sin \theta + 1 = \frac{-1}{\sqrt{R}} \sin \frac{\phi}{2}$$

$$\Rightarrow \left(\frac{1}{\sqrt{R}} \cos \frac{\phi}{2} \right)^2 + \left(1 + \frac{1}{\sqrt{R}} \sin \frac{\phi}{2} \right)^2 = r^2$$

$$\Rightarrow \frac{1}{R} + 1 + \frac{2}{\sqrt{R}} \sin \frac{\phi}{2} = r^2 \quad \dots (1)$$

Therefore the unit circle $|z| = r = 1$ in z -plane corresponds to the region of ω -

plane defined by $1 + \frac{1}{R} + \frac{2}{\sqrt{R}} \sin \frac{\phi}{2} = 1$

$$\text{or, } \frac{1}{\sqrt{R}} = -2 \sin \frac{\phi}{2}$$

$$\Rightarrow \frac{1}{\sqrt{R}} = 4 \sin^2 \frac{\phi}{2}$$

$$\text{or, } \frac{1}{R} = 2(1 - \cos \phi)$$

This represents a parabola in ω -plane. Next we examine about the correspondence between the interior of the circle and exterior of the parabola.

If we suppose $|z| = r < 1$, then the interior of the circle $|z| = 1$ corresponds to the region ω -plane defined by

$$1 + \frac{1}{R} + \frac{2}{\sqrt{R}} \sin \frac{\phi}{2} = r^2 < 1, \text{ by (1)}$$

$$\text{or, } \frac{1}{\sqrt{R}} < -2 \sin \frac{\theta}{2}$$

$$\text{or, } \frac{1}{R} < 2(1 - \cos \phi)$$

$$\text{or, } R > \frac{1}{2(1 - \cos \phi)}$$

This represents domain outside the parabola.

$$R = \frac{1}{2(1 - \cos \phi)},$$

$$\text{i.e., } \frac{1}{R} = 2(1 - \cos \phi)$$

This completes the proof.

2.13 Model Questions

1. Find the fixed points and the normal form of the following bilinear transformations.

(a) $\omega = \frac{z}{z-2},$

(b) $\omega = \frac{3iz+1}{z+i},$

(c) $\omega = \frac{z-1}{z+1}$

(d) $\omega = \frac{(2+i)z-2}{z+i},$

(e) $\omega = \frac{z}{2-z}$

Is any of these transformations hyperbolic, elliptic or parabolic ?

2. Find the bilinear transformation which maps the points $z = -2, 0, 2$ into the points $\omega = 0, i, -i$ respectively.
3. Find the bilinear transformation which maps the points $z = 0, -1, \infty$ into the points $\omega = -1, -2, -i, i$ respectively.
4. Find linear maps for $z = 0, -i, -1$ and corresponding values of ω are $\omega = i, 1, 0$.
5. If $z = 2\omega + \omega^2$, then show that $|\omega| = 1$ corresponds to a cardioid of z -plane.
6. Find bilinear transformation which maps $0, i, -i$ of z -plane to $1, -1, 0$ of ω -plane.

7. Prove that the relation $\omega = \frac{iz+z}{4z+i}$ transforms the real axis in the z -plane into

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a circle in the ω -plane. Find the centre and radius of the circle and the point in the z -plane which is mapped on the centre of the circle.

8. Show that the transformation $\omega = \frac{5-4z}{4z-2}$ transforms the circle $|z| = 1$ into a circle of radius unity in the w -plane and find the centre of this circle.

9. Discuss the application of the transformation $w = \frac{iz+1}{z+i}$ the areas in the z -plane which are respectively inside and outside the unit circles with centre at the origin.

Answers

1. (a) $k = \frac{1}{2}e^{i\pi}$, map is loxodromic

(b) $\frac{1}{\omega-i} = \frac{1}{z-i} = \frac{-i}{2}$ which is the normal form. The transformation is parabolic,

(c) $k = -i = e^{-i\frac{\pi}{2}}$, hence transformation is elliptic,

(d) $u = \sqrt{5} e^{i\theta}$ so that transformation is loxodromic.

(e) $k = \frac{1}{2} > 0$ and $u \neq 1$. Hence transformation is hyperbolic.

2. $\omega = i \left(\frac{2+z}{2-3z} \right)$, **3.** $\omega = \frac{iz-2}{2+z}$ **4.** $\omega = -\frac{i(z+i)}{z-1}$ **5.** $r = 2(1 + \cos\theta)$ which

represents cardioid. **6.** $\omega = \frac{z+i}{-3z+i}$, $4u \neq 0$ represents a bilinear transformation.

7. The real axis in z -plane is transformed on to a circle in ω plane whose centre is transformed on to a circle in ω plane whose centre is at $-\frac{7i}{8}$ and radius $= \frac{9}{8}$ and the centre of the circle corresponds to the point $z = \frac{i}{4}$ in z -plane.

8. Centre $\left(-\frac{1}{2}, 0\right)$ and of radius one in ω -plane.

9. If $v > 0$ i.e., upper of ω plane corresponds to $|z| > 1$ i.e., exterior of the circle $|z| = 1$.

If $v < 0$, i.e., lower half of ω plane corresponds to $|z| < 1$ i.e interior of the circle $|z| = 1$.

2.14. References :

1. Functions of a Complex Variable — Goyal & Gupta.
2. Complex Analysis — by J.B. Conway.
3. Complex Function Theory — D. Sarason.
4. Function of a Complex Variable — J.N. Sharma.
5. Functions of a Complex Variable — M.L. Khanna.

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